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 Studierichting: Informatica
 Oefeningenreeks 2F nr. 2

$$\begin{aligned}
 \cosh(3x) &= \cosh(x + 2x) \\
 &= \cosh(x) \cdot \cosh(2x) + \sinh(x) \cdot \sinh(2x) \\
 &= \cosh(x)(\cosh^2(x) + \sinh^2(x)) + \sinh(x)(2 \sinh(x) \cosh(x)) \\
 &= \cosh^3(x) + \cosh(x) \sinh^2(x) + 2 \sinh^2(x) \cosh(x) \\
 &= \cosh(x)(\cosh^2(x) + 3 \sinh^2(x)) \\
 &= \cosh(x)(\cosh^2(x) + 3(\cosh^2(x) - 1)) \\
 &= \cosh(x)(\cosh^2(x) + 3 \cosh^2(x) - 3) \\
 &= 4 \cosh^3(x) - 3 \cosh(x) \\
 \sinh(3x) &= \sinh(x + 2x) \\
 &= \sinh(x) \cosh(2x) + \cosh(x) \sinh(2x) \\
 &= \sinh(x)(\cosh^2(x) + \sinh^2(x)) + \cosh(x)(2 \sinh(x) \cosh(x)) \\
 &= \sinh(x) \cosh^2(x) + \sinh^3(x) + \cosh(x) 2 \sinh(x) \cosh(x) \\
 &= \sinh(x)(\cosh^2(x) + \sinh^2(x) + 2 \cosh^2(x)) \\
 &= \sinh(x)(3 \cosh^2(x) + \sinh^2(x)) \\
 &= \sinh(x)(3 \sinh^2(x) + 3 + \sinh^2(x)) \\
 &= 4 \sinh^3(x) + 3 \sinh(x) \\
 \tanh(3x) &= \tanh(x + 2x) \\
 &= \frac{\tanh(x) + \tanh(2x)}{1 + \tanh(x) \tanh(2x)} \\
 &= \frac{\tanh(x) + \frac{2 \tanh(x)}{1 + \tanh^2(x)}}{1 + \tanh(x) \cdot \left(\frac{2 \tanh(x)}{1 + \tanh^2(x)} \right)} \\
 &= \frac{\frac{\tanh(x) + \tanh^3(x) + 2 \tanh(x)}{1 + \tanh^2(x)}}{\frac{1 + \tanh^2(x) + 2 \tanh^2(x)}{1 + \tanh^2(x)}}
 \end{aligned}$$

$$\begin{aligned}
&= \frac{\tanh(x) + \tanh^3(x) + 2 \tanh(x)}{1 + \tanh^2(x) + 2 \tanh^2(x)} \\
&= \frac{\tanh^3(x) + 3 \tanh(x)}{1 + 3 \tanh^2(x)}
\end{aligned}$$

References